

Overview

This SQL script contains a collection of database queries and administrative operations for analyzing retail sales data from the dmart_analysis dataset. The file demonstrates various SQL operations including data exploration, aggregation, window functions, and user management.

Dataset

The primary table analyzed is dmart_analysis.'train (1)', which contains retail sales information with the following key columns:

- item_identifier / item_id - Unique product identifier
- item_type - Category of the product
- item_fat_content - Fat content classification (e.g., 'low fat')
- item_mrp - Maximum Retail Price
- item_weight - Product weight
- item_outlet_sales - Sales figures per outlet
- Outlet_Establishment_Year / outlet_year - Year the outlet was established
- outlet_type - Classification of the outlet

Script Sections

1. Initial Data Exploration

SQL

```
SELECT * FROM dmart_analysis.`train (1)`;
```

```
select count(*) from dmart_analysis.`train (1)`;
```

```
select distinct item_type from dmart_analysis.`train (1)`;
```

```
SELECT * FROM dmart_analysis.`train (1)` limit 10;
```

Basic queries to explore dataset structure, row count, unique item types, and sample records.

2. Aggregation Analysis

SQL

```
SELECT item_fat_content, AVG(item_mrp) FROM dmart_analysis.`train (1)` GROUP BY item_fat_content;
```

```
select item_type, avg(item_weight) from dmart_analysis.`train (1)` where item_weight<10  
group by item_type having avg(item_outlet_sales);
```

```
SELECT item_type, sum(Item_Outlet_Sales) FROM dmart_analysis.`train (1)` GROUP BY  
Item_type;
```

Queries aggregating data by fat content, item type, and weight with GROUP BY and HAVING clauses.

3. Filtering & Time-based Queries

SQL

```
select item_type from dmart_analysis.`train (1)` where Outlet_Establishment_Year < 1999;
```

Filters data based on outlet establishment year.

4. Schema Changes

SQL

```
RENAME TABLE dmart_analysis TO dmart_analysis_data;
```

```
alter table dmart_analysis_data rename column Outlet_Establishment_Year to outlet_year;
```

```
alter table dmart_analysis_data rename column item_identifier to item_id;
```

Table and column renaming operations for better naming conventions.

5. Advanced SQL Operations

Window Functions:

SQL

```
SELECT item_type, item_outlet_sales, outlet_type,  
       avg(item_outlet_sales) OVER (PARTITION BY outlet_type ORDER BY outlet_year) AS year  
FROM dmart_analysis_data;
```

Subqueries:

SQL

```
SELECT * FROM dmart_analysis_data WHERE item_mrp > (SELECT AVG(item_mrp) FROM  
dmart_analysis_data);
```

CASE Statements:

SQL

```
SELECT item_type, item_mrp,
```

```

CASE WHEN item_mrp > 100 THEN 'High'

      WHEN item_mrp BETWEEN 50 AND 80 THEN 'Medium'

      ELSE 'Low'

END AS mrp_category

FROM dmart_analysis_data;

```

6. User & Permission Management

SQL

```
create user shiva identified by "12345";
```

```
grant select on dmart_analysis_data to shiva;
```

```
revoke select on dmart_analysis_data from shiva;
```

Creates a user account, grants SELECT privileges, and then revokes them.

7. Transaction Control

SQL

```
savepoint sp1;
```

Sets a savepoint for transaction rollback purposes.

Common Queries

Purpose	Query
Count total records	SELECT COUNT(*) FROM dmart_analysis_data;
Low-fat items count	SELECT COUNT(*) FROM dmart_analysis_data WHERE item_fat_content = 'low fat';
High-sales items	SELECT COUNT(*) AS sales_count FROM dmart_analysis_data WHERE item_outlet_sales > 1000;
Price category distribution	Uses CASE statement for categorization
Average price by fat content	Groups by item_fat_content with AVG aggregation

Notes

- The table name was renamed from dmart_analysis to dmart_analysis_data during the script execution
- Column names were standardized (e.g., Outlet_Establishment_Year → outlet_year)
- Some queries may contain syntax variations depending on the database system (MySQL, PostgreSQL, etc.)
- The script includes both data analysis and database administration operations

Usage

Execute these queries in your database client against the dmart_analysis_data database to perform retail sales analysis, create user accounts, and manage permissions.